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Position-Specific Differences in Speed Profiles Among National Football League Scouting Combine Participants

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Abstract

Gillen, ZM. Position-specific differences in speed profiles among NFL scouting combine participants. *J Strength Cond Res* XX(X): 000–000, 2024—This study examined the relationships between speed profiles and athletic performance measurements from the National Football League (NFL) Scouting Combine, and position-specific differences in speed profiles. Subjects included NFL Scouting Combines participants with valid scores for the 40-yard dash, as well as the 10- and 20-yard splits of the 40-yard dash (n = 2,156). Subjects were divided into the following position groups: defensive backs (DBs, n = 421), defensive linemen (DL, n = 356), linebackers (LBs, n = 261), offensive linemen (OL, n = 354), running backs (RBs, n = 217), tight ends (TEs, n = 122), quarterbacks (QBs, n = 108), and wide receivers (WRs, n = 317). Performance measures included 40-yard dash time (with 10- and 20-yard split times), bench press repetitions to failure, vertical jump height, broad jump distance, pro-agility time, and L-cone drill time. The 40-yard dash and its splits calculated v_{max}, τ, and a_{max} to reflect speed profiles. One-way analysis of variances examined position-related differences. Pearson's product moment correlation coefficients examined relationships between performance measures and speed profile variables. Skill positions (DB and WR) had the greatest v_{max} and a_{max}, and lowest τ, followed by big skill positions (LB, RB, TE, QB), followed by DL, then OL with the lowest v_{max} and a_{max}, and highest τ. For skill and big skill positions, v_{max}, τ, and a_{max} exhibited greater relationships with combine measures, whereas OL generally had the lowest relationships. The position-specific differences in this study demonstrate potential areas of strength and weakness for certain positions. This may help guide strength and conditioning coaches desiring to improve position-specific speed and acceleration capabilities for American football players.

Key Words: American football, velocity, acceleration

Introduction

The National Football League (NFL) Scouting Combine serves as an annual event in which coaches, scouts, and team officials can assess potential NFL players. The overarching goal of the NFL Scouting Combine is to provide a comprehensive assessment of overall athletic ability, which should serve to provide at least some insight to the potential success of players in the NFL at their respective position. Athletes who participate in the combine complete simple assessments of height and body mass, the 40-yd dash (with split times at 10 and 20 yards) to assess maximal speed and acceleration, the bench press test to assess upper-body strength and endurance, the vertical and broad jump tests (VJ and BJ, respectively) to assess lower-body power, and the pro-agility and L-cone drills (PA and LC, respectively) to assess change-of-direction ability. Although it is common to simply examine the raw scores from each test, it has recently been suggested that further examination of speed profiles from the 40-yard dash and its associated split times may provide further insight regarding athletic ability in NFL Scouting Combine participants (2).

Because the goal of the NFL Scouting Combine is to assess potential talent for future NFL players, it is imperative to examine what these data mean from a performance perspective. One metric that has been suggested to provide potential predictive information for potential NFL players is linear speed (2,16,20). In the NFL Scouting Combine, linear speed is quantified with the 40-yard dash, with split times recorded at the 10- and 20-yard marks. Thus, the 40-yard dash and its associated splits can provide information on maximum velocity and acceleration capabilities. Because of this, previous studies have suggested examining sprint velocity profiles (2,3,12,14,18), which may provide greater information than the raw 40-yard dash and split times. This method was initially proposed by Furusawa et al. (5) and further expanded on by Greene (7) and Chelly and Denis (1) with the following calculation:

$$v(t) = v_{\max}\left(1 - e^{\frac{-t}{\tau}}\right)$$

where v_{max} is maximal velocity during the sprint and τ is the time constant that represents the ratio of v_{max} to initial acceleration (a_{max}). Theoretically, smaller τ values reflect a faster acceleration to a high percentage of v_{max}. Thus, an examination of v_{max}, τ, and a_{max} may provide further information regarding sprinting and acceleration capabilities of athletes.

Only 1 previous study has examined v_{max}, τ, and a_{max} from the 40-yard dash and associated splits in NFL Scouting Combine participants (2). Clark et al. (2) demonstrated that athletes with faster v_{max} yielded faster acceleration capabilities, demonstrating the importance of prioritizing maximal speed training for NFL Scouting Combine participants. Although Clark et al. (2) clearly

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demonstrated the importance of maximal speed, it is unclear if v_{max}, τ, and a_{max} exhibit relationships with other NFL Scouting Combine measurements, which may provide insight regarding the unique information gained from these calculations. Furthermore, recent research has demonstrated position-specific differences for NFL Scouting Combine measurements (6), suggesting that the player's specific position may influence the outcome measurements and interpretation of these data. Although it can be inferred that position-specific differences are likely to emerge, no previous studies have examined position-specific differences for v_{max}, τ, and a_{max}. Therefore, the purpose of this study was to quantify position-specific differences in v_{max}, τ, and a_{max}. Furthermore, another purpose of this study was to quantify the relationships between v_{max}, τ, and a_{max} and other athletic performance measurements from the NFL Scouting Combine to examine what insight these metrics may provide beyond simple raw scores commonly used from the NFL Scouting Combine.

Methods

Experimental Approach to the Problem

This study analyzed NFL Scouting Combine performance data from 2017, 2018, 2019, 2022, and 2023. Because of the COVID-19 pandemic, data from 2020 to 2021 were excluded. Because Fitzgerald and Jensen (4) have previously demonstrated significant changes over time for these metrics, only data from the most recent 5 years were used. These data were used to examine position-related differences in speed profiles and correlations among speed profile variables (v_{max}, τ, and a_{max}) and other combine performance measures. Therefore, the independent variable in this study was position group, whereas the dependent variables were 40-yard dash time (with 10- and 20-yard split times), v_{max}, τ, a_{max}, bench press repetitions to failure, VJ height, BJ distance, PA time, and LC drill time.

Subjects

Subjects included American football players who participated in the NFL Scouting Combines in 2017, 2018, 2019, 2022, and 2023 and had valid scores for the 40-yard dash, as well as the 10- and 20-yard splits of the 40-yard dash (n = 2,156). Subjects were then divided into the following position groups: defensive backs (DBs, n = 421), defensive linemen (DL, n = 356), linebackers (LBs, n = 261), offensive linemen (OL, n = 354), running backs (RBs, n = 217), tight ends (TEs, n = 122), quarterbacks (QBs, n = 108), and wide receivers (WRs, n = 317). The Mississippi State University Institutional Review Board (IRB) determined that the data were considered exempt because the data are publicly accessible (22).

Procedures

Anthropometrics. For height and body mass subjects were required to wear compression garments and no shoes or cleats. Height was recorded in centimeters, and body mass was recorded in kilograms.

40-Yard Dash and Speed Profiles. For the 40-yd dash, subjects began in a 3-point sprinters stance with their hand on the ground. Subjects then sprinted straight forward as fast as possible through the finish line marked 40 yards from the starting line. The test began when the subject's hand broke contact with the ground and ended when the subject passed through the finish line. Split times were also recorded at the 10- and 20-yd marks. Quantitative evaluations of speed profiles required calculations of v_{max}, τ, and a_{max}. Predicted maximum velocity (v_{max}) and acceleration patterns (τ) were taken as the slope and x-axis intercept (y = 0), respectively, for the linear relationship between distance (m) and time (s) using the 10-, 20-, and 40-yd split times (1,2,5,7). Initial acceleration (a_{max}) was then calculation by dividing v_{max} by τ (1,2,5,7). An example of this is presented in Figure 1.

Bench Press. For the bench press test, subjects completed as many repetitions as possible with 225 lbs (102.3 kg). A repetition was counted if the subject lowered the bar all way to the chest and pressed the bar until full elbow extension. Subjects were required to keep their head, shoulders, and lower back/buttocks in contact with the bench, and right and left foot in contact with the ground. For QBs, only 4 subjects completed this test; therefore, these 4 data points were excluded for position-based comparisons and correlations.

Vertical Jump. For the VJ, standing reach of each subject was taken to establish maximal reach and a baseline mark. Subjects then performed a rapid countermovement of self-selected depth, followed by a maximal VJ. Jump height was taken as the difference between maximal reach height and maximal jump height to the nearest 0.5 inch (1.27 cm).

Broad Jump. For the BJ, subjects began with their toes even with the starting line, performed a rapid countermovement of self-selected depth, followed by a maximal horizontal jump as far as possible. Subjects were required to land with the feet planted firmly on the ground. Jump distance was taken as the distance from the starting line to the subject's heel closest to the starting line to the nearest 1.0 inch (2.54 cm).

Pro-Agility. For the PA, subjects began in a 3-point stance facing forward with their hand on the ground. Subjects then turned 90° to the right and ran 5 yards (4.57 m), touched a line on the turf with their right hand, turned 180° to the left and ran 10 yards, touched a line on the turf with their left hand, and turned back 180° to the right and ran 5 yards through the starting line. The PA began when the subject's hand broke contact with the ground before the initial turn and ended when they ran through the starting line at the end.

L-Cone Drill. For the LC, subjects began in a 3-point stance facing forward with their hand on the ground. Subjects then sprinted forward to a cone 5 yards away (4.57 m), touched the cone with their right hand, turned 180° and sprinted back to the starting line, turned 180° and sprinted toward the second cone, once past the cone, turned 90° to the right and sprinted toward a cone 5 yards away, circled around the third cone in a counterclockwise movement, sprinted back around the second cone, and finished across the starting line. The LC began when the subject's hand broke contact with the ground and ended when they ran through the starting line at the end.

Statistical Analyses

Means and 95% confidence intervals were calculated for all dependent variables. One-way analyses of variance with post hoc Bonferroni's corrections were used to test for any differences for

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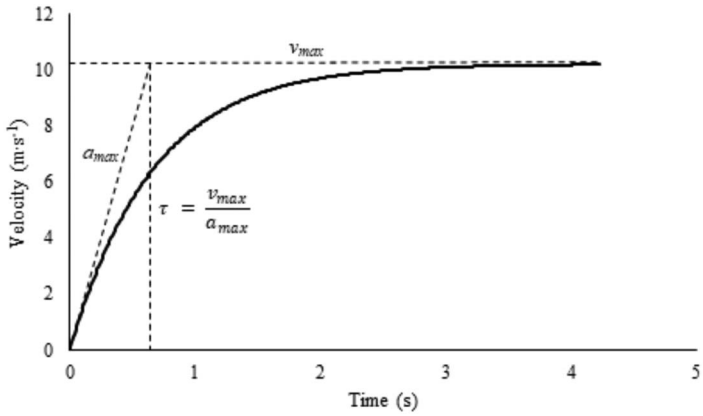

Figure 1. An example speed profile including v_{max}, τ, and a_{max}.

dependent variables across position groups. Pearson's product moment correlation coefficients were calculated to examine the relationships between combine performance measures and speed profile variables across the entire sample and split by position group. The following qualitative evaluations of the strength of association were made according to Mukaka (13) based on the absolute values of correlation coefficients: 0.90 to 1.00 = very high, 0.70 to 0.89 = high, 0.50 to 0.69 = moderate, 0.30 to 0.49 = low, and 0.00 to 0.29 = negligible. All statistical analyses were performed in IBM SPSS v. 28 (Chicago, IL). An alpha level of $p \leq 0.05$ was considered statistically significant.

Results

Means and 95% confidence intervals can be found in Table 1. There were significant position-related differences for all measurements: For height, OL, TE > DL > QB > LB > WR > DB > RB ($p \leq 0.016$). For body mass, OL > DL > TE > LB > QB, RB > WR, DB ($p < 0.001$). For 10- and 20-yd split, as well as 40-yd dash, DB, WR < RB < LB < TE, QB < DL < OL ($p \leq 0.003$). For v_{max}, DB, WR > RB > LB > TE, QB > DL > OL ($p \leq 0.002$). For τ, WR, DB < RB, LB, TE, QB, DL < OL ($p \leq 0.047$; note: WR, DB = RB, $p \geq 0.507$; LB < DL, $p < 0.001$). For a_{max}, DB, WR > RB, LB, TE, QB > DL, OL ($p \leq 0.033$; note: DB, WR = RB, $p \geq 0.721$; RB > QB, $p < 0.001$; QB = DL, $p = 0.151$). For bench press, OL, DL > LB > RB, TE > DB, WR ($p \leq 0.001$). For VJ, DB, WR > LB, RB, TE > QB > DL > OL ($p \leq 0.016$). For BJ, DB, WR > LB, RB, TE > QB, DL > OL ($p < 0.001$). For PA, DB, WR < RB, LB, TE, QB < DL < OL ($p < 0.001$). For LC, DB, WR < LB, RB, TE, QB < DL < OL ($p < 0.001$). Visual comparisons of the speed profiles for the fastest and slowest players for each position group are presented in Figure 2.

The relationships between v_{max}, τ, and a_{max} with all combine tests for the entire sample are in Table 2. There were low to very high significant relationships between v_{max} and combine tests ($|r| \geq 0.485$). There were negligible to moderate significant relationships between τ and combine tests ($|r| \geq 0.188$). There were negligible to moderate significant relationships between a_{max} and combine tests ($|r| \geq 0.269$).

The relationships between v_{max}, τ, and a_{max} with all combine tests split by position are in Table 3. For DBs, v_{max} and a_{max} exhibited negligible to very high relationships with 10- and 20-yd split, 40-yd dash, VJ, BJ, PA, and LC ($|r| \geq 0.098$), and τ exhibited negligible to high relationships with 10- and 20-yd split, 40-yd dash, VJ, BJ, and LC ($|r| \geq 0.122$). For DL, v_{max}, a_{max}, and τ exhibited negligible to very high relationships with height, body mass, 10- and 20-yd split, 40-yd dash, VJ, BJ, PA, and LC ($|r| \geq 0.154$). For LBs, v_{max} exhibited negligible to very high relationships with body mass, 10- and 20-yd split, 40-yd dash, VJ, BJ, PA, and LC ($|r| \geq 0.155$), whereas a_{max} and τ exhibited negligible to very high relationships with 10- and 20-yd split, 40-yd dash, bench press, VJ, BJ, PA, and LC ($|r| \geq 0.226$). For OL, v_{max} exhibited negligible to very high relationships with body mass, 10- and 20-yd split, 40-yd dash, VJ, BJ, PA, and LC ($|r| \geq 0.240$), τ exhibited negligible to high relationships with 10- and 20-yd split and VJ ($|r| \geq 0.108$), and a_{max} exhibited negligible to low relationships with 10- and 20-yd split, 40-yd dash, VJ, BJ, PA, and LC ($|r| \geq 0.180$). For RBs, v_{max} exhibited negligible to very high relationships with all combine assessments ($|r| \geq 0.160$), τ exhibited negligible to high relationships with height, body mass, 10- and 20-yd split, 40-yd dash, BJ, and LC ($|r| \geq 0.155$), and a_{max} exhibited negligible to very high relationships with height, body mass, 10- and 20-yd split, 40-yd dash, VJ, BJ, and LC ($|r| \geq 0.287$). For TEs, v_{max} exhibited negligible to very high relationships with height, 10- and 20-yd split, 40-yd dash, VJ, BJ, PA, and LC ($|r| \geq 0.179$), τ exhibited negligible to high relationships with height, 10- and 20-yd split, 40-yd dash, bench press, VJ, BJ, PA, and LC ($|r| \geq 0.187$), and a_{max} exhibited negligible to very high relationships with 10- and 20-yd split, 40-yd dash, bench press, VJ, BJ, PA, and LC ($|r| \geq 0.273$). For QBs, v_{max}, τ, and a_{max} exhibited negligible to very high relationships with 10- and 20-yd split, 40-yd dash, VJ, BJ, PA, and LC ($|r| \geq 0.212$). For WRs, v_{max} exhibited negligible to high relationships with 10-yd split, 40-yd dash, VJ, BJ, P, and LC ($|r| \geq 0.153$), τ exhibited negligible to high relationships with 10- and 20-yd split, VJ, and PA ($|r| \geq 0.114$), and a_{max} exhibited negligible to moderate relationships with 20-yd split, 40-yd dash, BJ, and PA ($|r| \geq 0.134$).

Discussion

The primary results of this study demonstrated position-specific differences for v_{max}, τ, and a_{max}, as well as position-specific relationships among v_{max}, τ, and a_{max} and other combine measures. Specifically, skill positions (DB and WR) tended to have the greatest v_{max} and a_{max} and lowest τ, followed by big skill positions (LB, RB, TE, QB), followed by DL, then OL with the lowest v_{max} and a_{max} and highest τ. Furthermore, for skill and big skill position groups, v_{max}, τ, and a_{max} tended to be related to other

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Table 1

Mean ± 95% confidence intervals for all performance measures by position group.

	Position group							
Variable	**DB**	**DL**	**LB**	**OL**	**RB**	**TE**	**QB**	**WR**
Height (cm)	182.49 ± 0.40†‡§‖¶#**	191.22 ± 0.37*‡§‖¶#**	186.74 ± 0.42*†§‖¶#**	194.53 ± 0.42*†‡‖#**	180.00 ± 0.62*†‡§¶#**	194.28 ± 0.60*†‡‖#**	189.59 ± 0.83*†‡§‖¶**	184.37 ± 0.65*†‡§‖¶#
Body mass (kg)	89.30 ± 0.45†‡§‖¶#	128.49 ± 0.27*‡§‖¶#**	107.23 ± 0.48*†§‖¶#**	140.44 ± 0.67*†‡‖¶#**	97.80 ± 1.02*†‡§¶**	113.02 ± 0.76*†‡§‖#**	99.43 ± 0.96*†‡§¶**	90.53 ± 0.79†‡§‖¶#
10-yd split (s)	1.57 ± 0.01†‡§‖¶#	1.72 ± 0.01*‡§‖¶#**	1.62 ± 0.01*†§‖¶#**	1.81 ± 0.01*†‡‖¶#,**	1.59 ± 0.01*†‡§ ¶#**	1.65 ± 0.01*†‡§‖**	1.67 ± 0.01*†‡§‖**	1.57 ± 0.01†‡§‖¶#
20-yd split (s)	2.62 ± 0.01†‡§‖¶#	2.86 ± 0.01*‡§‖¶#**	2.71 ± 0.01*†§‖¶#**	3.01 ± 0.01*†‡‖¶#**	2.66 ± 0.01*†‡§ ¶#**	2.75 ± 0.01*†‡§‖**	2.79 ± 0.02*†‡§‖**	2.61 ± 0.01†‡§‖¶#
40-yd dash (s)	4.52 ± 0.01†‡§‖¶#	4.96 ± 0.02*‡§‖¶#**	4.68 ± 0.02*†§‖¶#**	5.25 ± 0.02*†‡‖¶#**	4.59 ± 0.02*†‡§ ¶#**	4.76 ± 0.02*†‡§‖**	4.82 ± 0.03*†‡§‖**	4.52 ± 0.01†‡§‖¶#
v_{max} (m•s^{-1})	9.33 ± 0.02†‡§‖‖¶#	8.51 ± 0.04*‡§‖¶#**	9.02 ± 0.04*†§‖¶#**	8.00 ± 0.04*†‡‖¶#**	9.21 ± 0.04*†‡§ ¶#**	8.88 ± 0.05*†‡§‖**	8.75 ± 0.06*†‡§‖**	9.34 ± 0.03†‡§‖¶#
τ (s)	0.62 ± 0.01†‡§ ¶#	0.67 ± 0.01*‡§‖**	0.63 ± 0.01*†§**	0.69 ± 0.01*†‡‖¶#**	0.63 ± 0.01†§	0.65 ± 0.01*§**	0.65 ± 0.01*§**	0.61 ± 0.01†‡§ ¶#
a_{max} (m•s^{-2})	15.21 ± 0.12†‡§ ¶#	12.87 ± 0.14*‡§‖¶**	14.33 ± 0.16*†§**	11.44 ± 0.32*†‡‖¶#**	14.80 ± 0.17†§ ¶#	13.77 ± 0.21*†§‖**	13.56 ± 0.23*§‖**	14.93 ± 0.43†‡§¶#
Bench press repetitions	14.73 ± 0.43†‡§‖¶	23.95 ± 0.57*‡‖¶**	20.87 ± 0.6*†§‖¶**	24.19 ± 0.62*‡‖¶**	18.89 ± 0.76*†‡§**	18.58 ± 0.65*†‡§**	—	14.04 ± 0.55†‡§‖¶
Vertical jump height (cm)	89.71 ± 0.79†‡§‖¶#	77.49 ± 1.00*‡§‖¶**	86.46 ± 1.03*†§#**	70.34 ± 0.84*†‡‖¶#**	85.54 ± 1.08*†§#**	83.66 ± 1.45*†§#**	79.33 ± 1.60*‡§‖¶**	88.86 ± 0.87†‡§‖¶#
Broad jump distance (cm)	310.40 ± 1.56†‡§‖¶#	281.97 ± 2.32*‡§‖¶**	303.02 ± 1.9*†§#**	265.13 ± 1.83*†‡‖¶#**	301.96 ± 2.04*†§#**	298.31 ± 2.76*†§#**	286.82 ± 3.18*‡§‖¶**	310.22 ± 1.76†‡§‖¶#
Pro-agility (s)	4.27 ± 0.02†‡§‖¶#	4.59 ± 0.03*‡§‖¶#**	4.39 ± 0.02*†§**	4.78 ± 0.02*†‡‖¶#**	4.37 ± 0.02*†‡§**	4.42 ± 0.03*†§**	4.40 ± 0.03*†§**	4.30 ± 0.02†‡§‖¶#
L-cone drill (s)	6.99 ± 0.02†‡§‖¶#	7.50 ± 0.04*‡§‖¶#**	7.14 ± 0.03*†§**	7.84 ± 0.03*†‡‖¶#**	7.16 ± 0.04*†§**	7.20 ± 0.03*†§**	7.18 ± 0.05*†§**	7.02 ± 0.03†‡§‖¶#

*DB = defensive back.
†DL = defensive linemen.
‡LB = linebacker.
§OL = offensive linemen.
‖RB = running back.
¶TE = tight end.
#QB = quarterback.
**WR = wide receiver. Superscript numbers indicate significant differences with the associated position group previously listed ($p \leq 0.05$).

combine measures, whereas OL appeared to have the lowest relationships, with τ having the weakest relationships with other combine measures. Although v_{max}, τ, and a_{max} have been quantified from NFL Scouting Combine participants (2), this study is the first to provide position-specific comparisons, which may help strength and conditioning professionals better understand the role that speed and acceleration play for specific positions in American football. These results could potentially help guide the design and implementation of position-specific strength and conditioning programs for American football players.

Maximal speed and acceleration are important abilities in American football. In an analysis of work-to-rest ratios in American football, Iosia and Bishop (8) found that the average running play lasts 4.86 seconds, whereas in Division IA collegiate American football, the average running play lasts 5.60 seconds. Thus, short, high-intensity bouts of maximal effort sprints, such as the 40-yd dash and associated splits, may accurately help coaches and practitioners assess potential NFL players. Furthermore, in studying practice drills in American football, Mamon Jr. et al. (11) recently suggested that most positions, except DL and OL, have a high degree of high-speed running (>12 mph) intensities throughout the practice week. This study also reported that DL and OL tend to have lower high-speed running, but a larger volume of high acceleration intensities, which is in line with conclusions in previous studies (19,21). These results tend to agree with previous studies, with DB and WR achieving the highest v_{max} and a_{max} and lowest τ, and linemen achieving the lowest v_{max} and a_{max} and highest τ. Theoretically, τ represents how quickly an individual accelerates relative to maximum velocity (2). Thus, greater τ values reflect a runner taking longer to reach a high percentage of their maximum velocity. Therefore, the greater volume of maximum acceleration movements rather than maximum velocity movements in linemen could account for the higher τ as they are not typically accelerating completely to maximum velocity during practices or games. This is further supported by a recent study demonstrating that skill positions tend to have the greatest scores for jumping, change of direction, and sprinting assessments from the NFL Scouting Combine (6), suggesting that position-specific adaptations to the physiological demands of American football may largely account for the differences in v_{max}, τ, and a_{max}. However, it is important to note that positions who generally generate greater force (DL and OL) because of greater strength do benefit from greater overall body mass, even if the trade-off is lower velocity production and acceleration. For example, acceleration is equal to force divided by mass. Thus, even with greater force generation, as commonly seen in linemen, the larger body mass inherently decreases acceleration. Furthermore, many athletes are selected for a given position based on certain genetic factors that may predispose them to performing better at that position. For example, an individual with a greater body size based on genetics may be better suited to play DL or OL, as opposed to a smaller individual who may be

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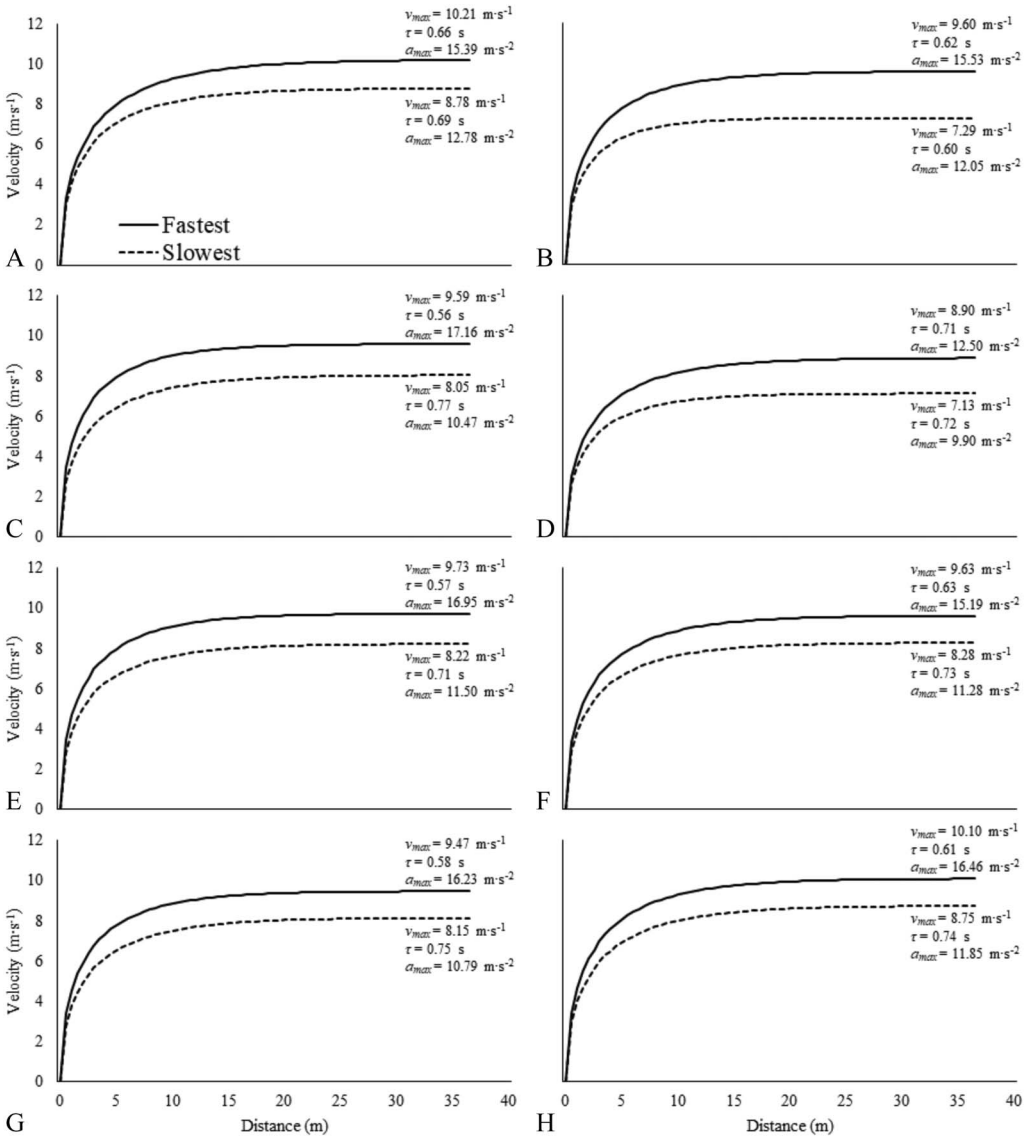

Figure 2. Visual comparisons of the speed profiles for the fastest and slowest players for (A) defensive backs, (B) defensive linemen, (C) linebackers, (D) offensive linemen, (E) running backs, (F) tight ends, (G) quarterbacks, and (H) wide receivers.

more suited to play a skill position such as WR or DB. This could lead to certain areas of strength and weakness for a given ability based on factors beyond position-specific adaptations.

Interestingly, among NFL Scouting Combine participants, Clark et al. (2) suggested that τ values generally remain similar across a heterogenous sample of runners, suggesting that position-specific differences in τ should be nominal. This hypothesis is mathematically supported from the calculation of τ as the ratio of v_{max} to a_{max} remains fairly constant even across a sample of runners with a large degree of variation in maximum speed. This can be further examined by the concept of "mass-specific force," where the ability to accelerate to maximum speed

Table 2

Pearson's product moment correlation coefficients for the relationship between speed profile variables (v_{max}, τ, and a_{max}), and height, body mass, 10- and 20-yd split times, 40-yd dash, bench press repetitions, vertical jump (VJ), broad jump (BJ), pro-agility (PA), and L-cone drill (LC) for the entire sample.

Combined	Height (n = 2,156)	Body mass (n = 2,156)	10-yd split (n = 2,156)	20-yd split (n = 2,156)	40-yd dash (n = 2,156)	Bench press (n = 1,648)	VJ (n = 2,077)	BJ (n = 2,052)	PA (n = 1,849)	LC (n = 1,802)
v_{max} (m•s^{-1})	−0.575*	−0.831*	−0.846*	−0.855*	−0.975*	−0.485*	0.714*	0.785*	−0.748*	−0.767*
τ (s)	0.261*	0.392*	0.634*	0.662*	0.415*	0.188*	−0.396*	−0.406*	0.320*	0.366*
a_{max} (m•s^{-2})	−0.348*	−0.511*	−0.678*	−0.429*	−0.600*	−0.269*	0.484*	0.535*	−0.481*	−0.504*

*Significant correlation ($p \leq 0.05$).

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Table 3

Pearson's product moment correlation coefficients for the relationship between speed profile variables (v_{max}, τ, and a_{max}), and height, body mass, 10- and 20-yd split times, 40-yd dash, bench press repetitions, vertical jump (VJ), broad jump (BJ), pro-agility (PA), and L-cone drill (LC) split by position group.

Defensive backs	Height (n = 421)	Body mass (n = 421)	10-yd split (n = 421)	20-yd split (n = 421)	40-yd dash (n = 421)	Bench press (n = 335)	VJ (n = 388)	BJ (n = 384)	PA (n = 341)	LC (n = 327)
v_{max} (m•s^{-1})	−0.038	−0.098*	−0.322*	−0.533*	−0.888*	0.018	0.275*	0.334*	−0.225*	−0.152*
τ (s)	−0.036	0.055	0.804*	0.627*	0.231*	−0.074	−0.213*	−0.238*	0.071	0.122*
a_{max} (m•s^{-2})	0.024	−0.089	−0.916*	−0.809*	−0.515*	0.074	0.309*	0.350*	−0.147*	−0.173*
Defensive linemen	**Height (n = 356)**	**Body Mass (n = 356)**	**10-yd split (n = 356)**	**20-yd split (n = 356)**	**40-yd dash (n = 356)**	**Bench press (n = 305)**	**VJ (n = 342)**	**BJ (n = 335)**	**PA (n = 304)**	**LC (n = 305)**
v_{max} (m•s^{-1})	0.230*	−0.667*	−0.758*	−0.873*	−0.969*	−0.057	0.665*	0.777*	−0.605*	−0.630*
τ (s)	−0.154*	0.372*	0.749*	0.594*	0.406*	0.095	−0.433*	−0.413*	0.317*	0.422*
a_{max} (m•s^{-2})	0.233*	−0.602*	−0.947*	−0.879*	−0.771*	−0.100	0.642*	0.676*	−0.529*	−0.633*
Linebackers	**Height (n = 261)**	**Body Mass (n = 261)**	**10-yd split (n = 261)**	**20-yd split (n = 261)**	**40-yd dash (n = 261)**	**Bench press (n = 204)**	**VJ (n = 257)**	**BJ (n = 254)**	**PA (n = 215)**	**LC (n = 215)**
v_{max} (m•s^{-1})	0.004	−0.155*	−0.560*	−0.768*	−0.932*	0.060	0.419*	0.502*	−0.316*	−0.328*
τ (s)	−0.089	−0.020	0.817*	0.611*	0.396*	−0.254*	−0.427*	−0.390*	0.226*	0.255*
a_{max} (m•s^{-2})	0.081	−0.044	−0.941*	−0.836*	−0.693*	0.255*	0.543*	0.532*	−0.319*	−0.348*
Offensive linemen	**Height (n = 354)**	**Body Mass (n = 354)**	**10-yd split (n = 354)**	**20-yd split (n = 354)**	**40-yd dash (n = 354)**	**Bench press (n = 296)**	**VJ (n = 348)**	**BJ (n = 346)**	**PA (n = 329)**	**LC (n = 321)**
v_{max} (m•s^{-1})	0.021	−0.244*	−0.587*	−0.371*	−0.921*	0.240*	0.421*	0.467*	−0.410*	−0.380*
τ (s)	0.020	0.019	0.367*	0.712*	0.049	−0.045	−0.104	−0.108*	0.072	0.088
a_{max} (m•s^{-2})	−0.038	−0.100	−0.430*	0.359*	−0.313*	0.114	0.180*	0.257*	−0.198*	−0.267*
Running backs	**Height (n = 217)**	**Body Mass (n = 217)**	**10-yd split (n = 217)**	**20-yd split (n = 217)**	**40-yd dash (n = 217)**	**Bench press (n = 166)**	**VJ (n = 215)**	**BJ (n = 204)**	**PA (n = 166)**	**LC (n = 159)**
v_{max} (m•s^{-1})	−0.338*	−0.516*	−0.540*	−0.734*	−0.935*	−0.160*	0.437*	0.545*	−0.331*	−0.308*
τ (s)	0.155*	0.254*	0.747*	0.534*	0.255*	0.007	−0.121	−0.175*	0.017	0.218*
a_{max} (m•s^{-2})	−0.287*	−0.435*	−0.913*	−0.791*	−0.599*	−0.067	0.271*	0.366*	−0.143	−0.321*
Tight ends	**Height (n = 122)**	**Body Mass (n = 122)**	**10-yd split (n = 122)**	**20-yd split (n = 122)**	**40-yd dash (n = 122)**	**Bench press (n = 111)**	**VJ (n = 119)**	**BJ (n = 119)**	**PA (n = 118)**	**LC (n = 115)**
v_{max} (m•s^{-1})	−0.179*	−0.119	−0.493*	−0.705*	−0.916*	0.184	0.334*	0.508*	−0.236*	−0.220*
τ (s)	−0.187*	0.008	0.820*	0.620*	0.362*	−0.230*	−0.354*	−0.384*	0.335*	0.258*
a_{max} (m•s^{-2})	0.118	−0.042	−0.945*	−0.828*	−0.654*	0.273*	0.452*	0.544*	−0.385*	−0.315*
Quarterbacks	**Height (n = 108)**	**Body Mass (n = 108)**	**10-yd split (n = 108)**	**20-yd split (n = 108)**	**40-yd dash (n = 108)**	**Bench press**	**VJ (n = 102)**	**BJ (n = 100)**	**PA (n = 100)**	**LC (n = 92)**
v_{max} (m•s^{-1})	−0.185	−0.163	−0.565*	−0.744*	−0.944*	—	0.412*	0.469*	−0.367*	−0.273*
τ (s)	0.084	−0.022	0.726*	0.542*	0.212*	—	−0.433*	−0.454*	0.222*	0.317*
a_{max} (m•s^{-2})	−0.148	−0.036	−0.919*	−0.815*	−0.578*	—	0.565*	0.607*	−0.346*	−0.395*
Wide receivers	**Height (n = 317)**	**Body mass (n = 317)**	**10-yd split (n = 317)**	**20-yd split (n = 317)**	**40-yd dash (n = 317)**	**Bench press (n = 227)**	**VJ (n = 306)**	**BJ (n = 310)**	**PA (n = 276)**	**LC (n = 268)**
v_{max} (m•s^{-1})	−0.078	−0.060	−0.165*	0.092	−0.778*	0.073	0.247*	0.291*	−0.294*	−0.153*
τ (s)	0.102	0.187*	0.567*	0.851*	0.106	−0.095	−0.114*	−0.105	−0.133*	−0.029
a_{max} (m•s^{-2})	0.022	0.034	−0.217*	0.590*	−0.134*	0.057	0.065	0.152*	−0.177*	−0.102

*Indicates a significant correlation ($p \leq 0.05$).

is dependent on the mass-specific horizontal force exerted in the ground (9,15). Expanding on this, theoretically, individuals with the greatest v_{max} results produce greater mass-specific vertical force because at maximum velocity, most forces are downward (2). Therefore, the implication of this hypothesis is a constant ratio of v_{max} to a_{max}, yielding similar τ values across individuals of different running speeds (2). However, this study found results contradicting this hypothesis, such that the position groups with the greatest v_{max} results also yielded the greatest a_{max} results and lowest τ results. This may further lend support to the idea of position-specific adaptations because of practices and games, such that positions that regularly accelerate up to maximum speed (DBs and WRs) will not only have higher v_{max} results but also have higher a_{max} results with lower τ results, whereas positions that do not regularly accelerate up to maximum speed will not only have worse v_{max} results but alo have subsequently worse a_{max} and τ results. This also lends support to the idea that individuals with certain genetic factors, such as greater body size for individuals selected to play DL and OL, may necessarily have worse v_{max}, a_{max}, and τ. Thus, future studies should consider examining whether position-specific adaptations or genetic factors play a greater role in position-specific differences in speed profiles. Nevertheless, no previous studies have examined position-specific differences for speed profiles, indicating the need of future studies to expand upon these findings.

Furthermore, these results may reflect areas of weakness for certain positions, potentially highlighting outcome variables that could be prioritized for position-specific training programs for American football players. For instance, an athlete may have position-specific adaptations or different genetic factors

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yielding worse acceleration and sprinting abilities compared with other positions. That is not to say that those abilities should overlooked in a training program; in fact, it may benefit a strength and conditioning coach to prioritize improving those abilities to gain a competitive edge over other athletes of the same position on an opposing team. To demonstrate this, Figure 2 presents a visual comparison of the speed profiles for the fastest and slowest players within each position. From a visual inspection, the shapes of the curve for the fastest and slowest players appear similar. However, as can be seen from the data provided in each panel, v_{max}, τ, and a_{max} appear to differ somewhat greatly. From a training perspective, this has practical implications. For example, although 2 athletes of the same position may have similar speed profiles, a strength and conditioning coach could emphasize maximal acceleration and speed training to improve v_{max}, τ, and a_{max}. Using DL and OL as an example, many strength and conditioning coaches may take the results of previous studies (6,11,19,21) and conclude that they need to emphasize maximal acceleration training with short yardage bursts of activity, although maximal speed training, involving longer sprints, may not be as great of a priority. However, it is possible performing both in a properly designed strength and conditioning program, particularly because of the physiological adaptations in high-threshold motor units that accompany maximal speed training, should theoretically yield greater improvements in speed and acceleration capabilities. This has been previously explained by Ross et al. (17) who suggested that sprint training, such as sprinting distances between 60 and 100 m, can yield significant improvements in neuromuscular function, such as increases in nerve conduction velocity, improved stretch reflex, and improved neuromuscular efficiency, all of which lend to improve short-burst, high-intensity acceleration movements. Furthermore, in examining 6-second sprint intervals on a cycle ergometer (which would equate to longer, linear sprints in a field-based setting), Lanzi et al. (10) suggested that sprint interval training with short rest periods of 30 seconds, similar to rest periods between plays in American football, may yield significant metabolic improvements and alter motor unit recruitment strategies, yielding better recovery between high-intensity movement bouts. Therefore, it may be pertinent to train in such a way to maximize acceleration and speed, not just one or the other, to enhance overall performance and potentially recoverability between plays among American football athletes.

Regarding the correlations, v_{max}, τ, and a_{max} and were related to the 10- and 20-yd splits, as well as 40-yd dash, which is unsurprising given that the mathematical calculation for v_{max}, τ, and a_{max} involves these metrics. These results are in agreement with Clark et al. (2) who reported similar results when examining these relationships. However, no previous study has examined the relationship between v_{max}, τ, and a_{max} and the other NFL Scouting Combine tests. This study demonstrated that v_{max}, τ, and a_{max} exhibited significant relationships with all other combine metrics. When split by position, many of these relationships remained significant, although with lower magnitudes of the relationship. Perhaps one of the most unique findings regrding position-specific relationships was the greater relationships (in magnitude and significance) between anthropometrics (height and body mass) and v_{max}, τ, and a_{max} in DL, OL, and RBs. As previously mentioned, DL and OL generally benefit from greater body mass because of the nature of their position. Furthermore, RBs can benefit from greater body mass because of the collision nature of their position, such that it is common for RBs to initially run against DL for the first few yards during a running play. Nevertheless, among these positions, there were negative relationships between anthropometrics and v_{max}, τ, and a_{max}. Although one might conclude that decreasing body mass could yield better speed profiles, this may be a poor trade-off for these positions, which speaks to the complexity of the relationship between speed, acceleration, and body mass as faster DL and OL may not necessarily meet the position-specific demands of heavier DL and OL, although more research is needed regarding this. For all other combine tests across positions, τ and a_{max} generally exhibited lower magnitudes of relationship with other combine tests than v_{max}. Although one may conclude that this diminishes the value of τ and a_{max}; statistically speaking, this provides insight that these metrics may provide unique information not gathered by the other athletic performance tests. The goal of a testing battery in athletics is to examine the requisite abilities specific to a given sport. When all variables are significantly related, this might suggest concurrent validity, demonstrating redundancy among certain tests. Of note, τ generally exhibited lower relationships with the 40-yd dash across all positions. Thus, although the 40-yd dash and associated splits are necessary to calculate τ, it may provide unique physiological or biomechanical information not garnered by simply assessing split times. To provide greater insight into potential on-field performance, individuals evaluating NFL Scouting Combine participants, as well as American football players, may benefit from calculating v_{max}, τ, and a_{max}.

In conclusion, this study provides unique, position-specific comparisons of v_{max}, τ, and a_{max}, building off previous studies examining these metrics in NFL Scouting Combine participants (2). Overall, skill positions (DB and WR) had the greatest v_{max} and a_{max}, and lowest τ, followed by big skill positions (LB, RB, TE, QB), followed by DL, then OL with the lowest v_{max} and a_{max}, and highest τ. Furthermore, based on the correlations, it is possible that v_{max}, τ, and a_{max} may provide unique information not gathered in the other combine tests, although further research regarding this is necessary. This information may be useful to strength and conditioning coaches who wish to maximize athletic performance among American football players, particularly when considering the importance of maximal speed and acceleration to on-field performance.

Practical Applications

Although the goal of athletic performance combines is to assess the potential for on-field performance for a given sport, simply assessing the raw outcome variables may yield incomplete information. Specifically, in American football, maximal speed and acceleration are vital requisite performance characteristics that ultimately aid on-field performance. Therefore, calculating v_{max}, τ, and a_{max} from the 40-yd dash and associated splits may provide practical relevance for strength and conditioning coaches regarding sprinting and acceleration capabilities. Furthermore, the position-specific differences in this study demonstrate potential areas of strength and weakness for certain positions. Specifically, it appears that positions who are generally faster have greater acceleration profiles and achieve a higher percentage of their maximum velocity in a shorter amount of time. This may speak to the need of strength and conditioning coaches to prioritize training maximum speed, acceleration, or both depending on the results of the specific team.

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